$$\frac{A'B'}{PD} = \frac{A'F}{PF}$$

As, $PD = AB$ the above equation becomes,

$$\frac{A'B'}{AB} = \frac{A'F}{PF} \qquad (6.6)$$

From equations (6.5) and (6.6) we can write,

$$\frac{PA'}{PA} = \frac{A'F}{PF}$$

As, $A'F = PA' - PF$, the above equation becomes,

$$\frac{PA'}{PA} = \frac{PA' - PF}{PF} \qquad (6.7)$$

We can apply the sign conventions for the various distances in the above equation.

$PA = -u, \qquad PA' = -v, \qquad PF = -f$

All the three distances are negative as per sign convention, because they are measured to the left of the pole. Now, the equation (6.7) becomes,

$$\frac{-v}{-u} = \frac{-v-(-f)}{-f}$$

On further simplification,

$$\frac{v}{u} = \frac{v-f}{f}; \qquad \frac{v}{u} = \frac{v}{f} - 1$$

Dividing both sides with v,

$$\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$$

After rearranging,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \qquad (6.8)$$

The equation (6.8) is called ***mirror equation***. Although this equation is derived for a special situation shown in Figure (6.12), it is also valid for all other situations with any spherical mirror. This is because proper sign convention is followed for u, v and f in equation (6.7).

6.2.6 Lateral magnification in spherical mirrors

The *lateral* (or) *transverse* magnification *m* is defined as the ratio of the height of the image to the height of the object. The height of the object and image are measured perpendicular to the principal axis.

$$\text{magnification } (m) = \frac{\text{height of the image } (h')}{\text{height of the object } (h)}$$

$$m = \frac{h'}{h} \qquad (6.9)$$

Applying proper sign conventions for equation (6.5),

$$\frac{A'B'}{AB} = \frac{PA'}{PA}$$

$A'B' = -h', \text{AB} = \text{h}, PA' = -v, PA = -u$

$$\frac{-h'}{h} = \frac{-v}{-u}$$

On simplifying we get,

$$m = \frac{h'}{h} = -\frac{v}{u} \qquad (6.10)$$

Using mirror equation, we can further write the magnification as,

$$m = \frac{h'}{h} = \frac{f-v}{f} = \frac{f}{f-u} \qquad (6.11)$$

Note

The students are advised to refresh themselves with the image tracing for the concave and convex mirrors for various predetermined positions of the object; the position of image, nature of image etc., from 9th Standard, Science Book .

EXAMPLE 6.3

An object is placed at a distance of 20.0 cm from a concave mirror of focal length 15.0 cm.

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